

VIJEET GHATGE | +919833978804, +1 540-934-8780 | vijeet.financeusa@gmail.com | Boston (MA), Mumbai (India) | CFA Level II FRM Part II Candidate | **MS Mathematical Finance**

EDUCATION

MSF (Mathematical Finance) (STEM) | Babson College — Boston, MA | *Sep 2024 – May 2025*

Courses Taken: Derivatives Pricing, Trading Strategies, ML in Python & R, Economics & Financial Forecasting

B.Com. (Accounting & Auditing) | Mithibai College – Mumbai, India | *Jun 2017 – May 2020*

EXPERIENCE

Independent Derivatives Trader | *2022 – Present*

- **Long-term holding portfolio created in 2023 earned 93.889% return till date**
- **Intraday strategy earned Rs800000 on Rs200000 capital deployed using options**
- Computed and visualized weekly & monthly implied volatilities using BSM model & compared them with realized volatility to detect deviations in the vol-term structure & using these signals to structure spreads trades.

Intern – Finance & Accounting | Nehal R. Shah & Co – Mumbai | *Oct 2021 – Nov 2022*

- Benchmarked 10 mutual funds vs. indices; built audit models for 8+ firms/NGOs
 - Integrated IFRS/Ind AS frameworks across HNI portfolios
-

QUANT PROJECTS

- **Momentum Tracking Strategy**
 - Win rate across coins (58%) Alpha@5% SOL, XRP, AVAX (0.00526,0.00251,0.00443) resp
 - Designed and back tested a trading strategy **combining Dual Hull Moving Averages and Z-score reversion** across nine Layer 1 crypto assets.
- **Mean Reverting Pair Trade Strategy**
 - Win rate across coins (56%) generates a non-statistically significant alpha
 - Designed a **mean reversion pair trade strategy on Layer 1 crypto pairs using z-score-normalized spreads** and OLS-based hedge ratios, with entry filters based on **cointegration**
- **Developed a VWAP Algorithm for Liability Trading (LT3& LT4)**
 - **Designed & tested a VWAP algorithm** for executing a **100,000 - 1,000,000 share order** by analyzing intraday volume and price patterns to minimize impact cost
- **Simulated \$15M multi-asset portfolio as a Portfolio Manager & earned +22% ROI in 45 days by**

RESEARCH PROJECTS

- **Macroeconomic Sensitivity of Consumer Sector Equities:**
 - Modelled heterogeneous return responses to macroeconomic shocks of interest rate across entities classified based on gross margin tiers using **regression model** confirming rate exposure asymmetry between low and high margin firms with an **R² of 0.62**
 - **ARIMA Forecasting Henry Hub Gas Prices**
 - Modelled short term persistence via **ARIMA (1,0,1)** to identify statistically grounded hedge entries and coverage levels for NG
 - Forecasted range for last week of Jan25 (\$2.12 - \$4.07) MMBtu accurately contained realized prices
 - **Build a model to detect Volatility Spillover from SPX to Nifty**
 - Build a spillover detection engine to quantify how SPX and VIX volatility transmits into Nifty
 - **Used HAR – RV and GARCH and ML for future one day RV prediction**
 - **Hybrid ML model training RMSE = 2.146, test RMSE = 2.495**
-

SKILLS, TOOLS & Certificates

Tech: Python (Pandas & Numpy), R, Bloomberg (Market Concepts Certified), Excel C++(ongoing)

Finance: Derivatives Pricing, BSM, Binomial Tree, VaR modelling, **Interest Rate Tree Generation**

Econometrics: Linear Models (Gauss – Markov theorem), **Time Series Modelling & ARIMA models**

Soft: Analytical Thinking, Problem Solving, Story Telling, Ethics, Leadership

Certificates: BSE - Program on Derivatives & Investment Banking Certificate, **Wall Street Quant** – Quantitative Finance Bootcamp Certificate

My work – <https://vijeetghatge.github.io>